

iRecharge Platform

API Reference Manual

Version 1.2.0 • Enterprise Gateway Integration

Status: Production Active

Release Date: May 2026

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1. Introduction & Usage Flow

The iRecharge API suite allows high-speed automated prepaid/postpaid mobile recharges, DTH top-ups, utility bill payments, and money payouts. Our RESTful JSON APIs are designed to scale with your transaction load. The standard workflow follows simple steps: check your balance, query operator and circle plans, post a recharge transaction, and optionally receive transaction updates via our real-time webhook callbacks.

2. API Gateways & Environments

Gateway Environment	Base URL Address	Target Purpose
Production Gateway	https://rchserver.irecharge.in/api/v1	Live traffic & transactions
Sandbox Gateway	https://rchserver.irecharge.in/api/v1/sandbox	Integration testing and dry-runs

3. Authentication & Headers

All API requests must be transmitted securely over HTTPS and authenticated. Credentials are generated inside your iRecharge console under the security settings panel. Authentication is validated using API keys and/or Bearer Tokens. The following custom HTTP headers must accompany every API call:

Header Parameter	Type	Description & Example Value
Authorization	String	Bearer auth token: Bearer token_xyz123...
x-client-id	String	Your unique client identifier: client_usr_9a87d
x-api-key	String	Vetted dashboard API Key: ak_irecharge_29f7...
x-idempotency-key	String	A unique UUID per mutation request to prevent double-charges.

4. Core Recharge APIs

Submit a prepaid topup or postpaid payment request using your wallet balance.

HTTP Method	POST
Request Endpoint	/recharge
Content-Type	application/json

```
// REQUEST BODY PAYLOAD
{
  "mobile": "9876543210",
  "operatorCode": "JIO",
  "amount": 299,
  "circle": "MH"
}

// SUCCESS RESPONSE (200 OK)
{
  "success": true,
  "message": "Recharge initiated successfully",
  "data": {
    "transactionId": "TXN_987612345",
    "status": "PENDING",
    "operatorRef": "JIO-REF-89102",
    "amount": 299,
    "walletBalance": 2450.50
  }
}
```

5. Wallet & Payout APIs

5.1 Query Wallet Balance

HTTP Method	GET
Request Endpoint	/wallet/balance

```
// RESPONSE PAYLOAD (200 OK)
{
  "success": true,
  "data": {
    "balance": 2749.50,
    "currency": "INR",
    "cashbackBalance": 120.00
  }
}
```

5.2 Execute Payout (Direct Money Transfer)

HTTP Method	POST
Request Endpoint	/payout/transfer

```
// REQUEST BODY PAYLOAD
{
  "accountNumber": "9182736450",
  "ifsc": "UTIB0000234",
  "amount": 1500,
  "beneficiaryName": "John Doe",
  "payoutType": "IMPS"
}

// RESPONSE PAYLOAD (200 OK)
{
  "success": true,
  "message": "Payout dispatched successfully",
  "data": {
    "payoutId": "PAY_009988A",
    "status": "SUCCESS",
    "bankRrn": "659820120199",
    "fee": 5.00
  }
}
```

6. Operator & Circle APIs

Identify operators and fetch live tariff plans associated with a telephone number.

HTTP Method	GET
Request Endpoint	/operator/detect/{mobile}

```
// SUCCESS RESPONSE (200 OK)
{
  "success": true,
  "data": {
    "operatorCode": "JIO",
    "operatorName": "Reliance Jio",
    "circleCode": "MH",
    "circleName": "Maharashtra"
  }
}
```

7. Transaction Status APIs

Retrieve detailed operational stats of any historical transaction via unique identifier.

HTTP Method	GET
Request Endpoint	/transaction/status/{transactionId}

```
// SUCCESS RESPONSE (200 OK)
{
  "success": true,
  "data": {
    "transactionId": "TXN_987612345",
    "status": "SUCCESS",
    "operatorRef": "JIO-REF-89102",
    "mobile": "9876543210",
    "amount": 299.00,
    "timestamp": "2026-05-26T07:30:15.000Z"
  }
}
```

8. Webhooks & Callback Documentation

To avoid polling the status checking API repeatedly, configure your callback URL. The iRecharge platform issues a POST request to your specified callback endpoint immediately when the transaction status transitions to a final state (SUCCESS or FAILED).

```
// POST CALLBACK JSON PAYLOAD
{
  "event": "recharge.completed",
  "timestamp": 1779770000000,
  "data": {
    "transactionId": "TXN_987612345",
    "mobile": "9876543210",
    "amount": 299.00,
    "status": "SUCCESS",
    "operatorRef": "JIO-REF-89102",
    "errorMessage": null
  }
}
```

9. Integration SDK Code Examples

9.1 cURL Integration

```
curl -X POST https://rchserver.irecharge.in/api/v1/recharge \  
-H "Content-Type: application/json" \  
-H "Authorization: Bearer YOUR_TOKEN" \  
-H "x-client-id: YOUR_CLIENT_ID" \  
-H "x-api-key: YOUR_API_KEY" \  
-d '{"mobile": "9876543210", "operatorCode": "JIO", "amount": 299}'
```

9.2 Node.js Axios Example

```
const axios = require('axios');  
  
async function runRecharge() {  
  try {  
    const response = await axios.post('https://rchserver.irecharge.in/api/v1/recharge', {  
      mobile: '9876543210',  
      operatorCode: 'JIO',  
      amount: 299  
    }, {  
      headers: {  
        'Authorization': 'Bearer YOUR_TOKEN',  
        'x-client-id': 'YOUR_CLIENT_ID',  
        'x-api-key': 'YOUR_API_KEY'  
      }  
    });  
    console.log('Success:', response.data);  
  } catch (err) {  
    console.error('Error:', err.response?.data || err.message);  
  }  
}
```

9.3 PHP Native Example

```
<?php
$curl = curl_init();
curl_setopt_array($curl, [
    CURLOPT_URL => "https://rchserver.irecharge.in/api/v1/recharge",
    CURLOPT_RETURNTRANSFER => true,
    CURLOPT_POST => true,
    CURLOPT_POSTFIELDS => json_encode([
        "mobile" => "9876543210",
        "operatorCode" => "JIO",
        "amount" => 299
    ]),
    CURLOPT_HTTPHEADER => [
        "Content-Type: application/json",
        "Authorization: Bearer YOUR_TOKEN",
        "x-client-id: YOUR_CLIENT_ID",
        "x-api-key: YOUR_API_KEY"
    ]
]);
$response = curl_exec($curl);
curl_close($curl);
echo $response;
?>
```

10. Error Codes & Rate Limits

Code	HTTP Status	Meaning & Resolution Strategy
INVALID_CREDENTIALS	401 Unauthorized	Verify header keys and client authentication.
INSUFFICIENT_FUNDS	400 Bad Request	Top up your client wallet balance to transact.
OPERATOR_OFFLINE	503 Service Temp	The telecom downstream operator gateway is down.
RATE_LIMIT_EXCEEDED	429 Too Many Req	Limit is 60 req/min. Cool off for 1 minute.

11. Security Best Practices & Support

- 1. Server IP Whitelisting:** Restrict requests utilizing your API Keys strictly to your application backend servers.
- 2. Key Rotation:** Routinely cycle credentials in the developer portal console every 90 days.
- 3. HTTPS Enforcement:** Reject non-TLS/non-HTTPS local callbacks or API calls. Enforce TLS 1.2 or higher for handshake verification.
- 4. Double Transaction Prevention:** Ensure that the custom header x-idempotency-key transmits a unique hash per transaction. Retrying transactions using the same hash is safely ignored by the API gateways.

Developer & Operations Support

For critical production errors, API disruptions, or sandbox validation inquiries, contact the Developer Relations Group by emailing support at api-support@irecharge.in. Live API node health metrics and system maintenance windows are posted directly inside your operator merchant console.